

Atharva Chaudhari

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EDUCATION

University of Illinois Urbana-Champaign (UIUC) <i>MS in Information Management (August 2024 – May 2026)</i>	Champaign, IL <i>GPA: 4.0/4.0</i>
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EXPERIENCE

School of Information Science, UIUC NLP Engineer	Jan 2026 – Present
- Fine-tuned PubMedBERT on imbalanced biomedical data, achieving 91% F1 for clinical trial sample size extraction - Improved minority-class detection by 18% via resampling and custom weighted loss functions, reducing model bias and boosting fairness across underrepresented clinical populations - Leveraged BioMistral LLM for structured sample size extraction from biomedical literature via few-shot prompting - Optimized GPU inference using FP8 quantization and KV cache optimization, improving throughput by 31% on an NVIDIA H200 NVL - Built a token-level NER model extracting participant counts from unstructured biomedical text with 89% token accuracy	
Center for Health Informatics, UIUC AI/ML Engineer	Jul 2025 – Dec 2025
- Deployed a Multimodal LLM pipeline for real-time misinformation detection in videos using Whisper transcription, maintaining 99%+ uptime via OpenAI and Azure model fallbacks - Shipped a CI/CD-enabled video chunking pipeline with adaptive concurrent batching and a JSON-based prompt parser standardizing inputs across multiple LLM providers for production use	
Adarsh College of Computer and Management, Akola CS Instructor	Jun 2023 – May 2024
Taught C programming, Data Structures, Operating Systems, and Neural Networks to 200+ undergraduates	

PROJECTS

Adaptive Reasoning Budget Allocation via Reinforcement Learning for LLMs Link	Jan 2026 – Apr 2026
- Trained a PPO policy network on Qwen2.5 7B to dynamically allocate reasoning token budgets based on question difficulty, replacing the fixed budget paradigm used by standard LLMs - Designed a custom PPO reward function balancing reasoning accuracy and token efficiency, stabilized using value network pretraining and supervised fine-tuning - Reduced training label generation on NVIDIA H200 NVL from 2,100 sequential GPU calls to 7 batched CUDA kernel launches, cutting setup time from 3 hours to 21 minutes - Achieved 80% accuracy at 185 avg tokens vs 79% at 224 fixed budget tokens, saving 39 tokens per request without accuracy loss	
CareLens – Medical Query Classification Link	Aug 2025 – Dec 2025
- Built NLP system classifying patient queries into 10 disease categories for automated clinical triage - Curated synthetic training data using LLM prompting validated by medical students, ensuring clinical accuracy across disease categories - Benchmarked TF-IDF + LR/SVM/XGBoost against BERT, PubMedBERT, and Sentence-BERT; demonstrated SBERT generalizes 2x better on symptom-based out-of-distribution queries - Designed adversarial Lexical Scrubbing red teaming tests using statistical analysis to detect and eliminate shortcut learning, characterizing model failure modes for real-world robustness	
Dr. BOT – Healthcare NLP Chatbot UIUC Dr. BOT Challenge Second Prize Winner Link	Jun 2025 – Jul 2025
- Built a healthcare AI agent using LangChain that answers patient medical queries with clinical accuracy and emotional alignment - Fine-tuned PubMedBERT on 12,000+ MedQuAD QA pairs for disease domain classification, powering a domain-aware RAG pipeline that triggers targeted Wikipedia API retrieval for verified real-time clinical information - Integrated BioMistral for response generation ensuring factual accuracy, clinical reliability, and emotional alignment in the answer	
Skin Cancer Classification and Detection using VGG-19 and DenseNet Link	Jan 2023 – May 2023
- Built Deep Learning pipeline for dermatoscopic skin lesion segmentation using BCDU-Net, achieving Dice Coefficient of 90.66% and IOU of 83.09% on HAM10000 dataset - Trained and compared VGG-19 and DenseNet121 CNN architectures using transfer learning; VGG-19 achieved 97.29% accuracy, 97.42% precision, and 97.42% F1-score, outperforming DenseNet121 across all metrics Publication: International Conference on Computational Intelligence, Network and Security (IEEE) Link	
Household Furniture Detection for Visually Impaired People Link	Aug 2021 – Dec 2021
- Built a real time computer vision based assistive system converting indoor furniture detection into audio notifications for visually impaired users using SIFT feature extraction and four classifiers: SVM, Decision Tree, Random Forest, and KNN - Evaluated across accuracy, precision, recall, F1-score, sensitivity, and specificity; Decision Tree achieved 77.64% accuracy outperforming other classifiers Publication: International Conference on Intelligent Computing (IEEE) Link	

SKILLS

Programming Languages: Python, C++, C, JavaScript, SQL, x86 Assembly
Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Keras, Transformers
LLMs & NLP: RAG, LangChain, LlamaIndex, MCP, Prompt Engineering, Fine-Tuning, BERT, PubMedBERT, Sentence-BERT, Whisper
GPU Computing: CUDA 12.8, FP8 Quantization, KV Cache Optimization, Batched Inference
Computer Vision: OpenCV, CNNs, Transfer Learning, Image Segmentation, VGG-19, DenseNet
Cloud: AWS (EC2, S3, Lambda, RDS, IAM, VPC, ELB, Auto Scaling, CloudWatch, Route 53, EBS)
Certifications: AWS Certified Cloud Practitioner Credential